

Figure 1: Variant A: a fixed multiscale source is transported by a beam-conditioned Fourier velocity field.

1 No-Observation Operator Flow Matching

Let H denote the normalized building-height map, T the transmitter-location indicator, B the target beam map, and $a = (n, f, \theta)$ the known array configuration. The target y is a normalized radio-power map on the model grid. During training, y and the validity mask M provide supervision; neither the target map nor a same-scene auxiliary radio map is an inference input. We study two implementations, A and C, under the same flow-matching backbone and target data protocol.

1.1 Variant A: Beam-Conditioned Velocity with a Fixed Source

The A model uses the condition $c_A = [T, H, B, 0, 0, 0]$. Its source is a zero-mean multiscale Matérn random field, independent of the scene and beam. For spatial separation r, the component covariance is

$$C_\ell(r) = \begin{cases} (r/\ell)K_1(r/\ell), & r > 0, \\ 1, & r = 0, \end{cases} \qquad C_{\mathrm{MS}} = \frac{1}{3} \sum_{\ell \in \{2,8,32\}} C_\ell, \tag{1}$$

where K_1 is the modified Bessel function of the second kind and length scales are in model-grid pixels. The sampler uses a positive-semidefinite checked circulant embedding. In the paired A/C implementation, independent $z_{\mathrm{MS}} \sim \mathcal{GP}(0, C_{\mathrm{MS}})$ and $z_\ell \sim \mathcal{GP}(0, C_\ell)$ yield

$$x_0^A = \sqrt{0.75}\, z_{\mathrm{MS}} + \frac{0.5}{\sqrt{3}} \sum_{\ell \in \{2,8,32\}} z_\ell. \tag{2}$$

Thus x_0^A has covariance C_{MS}. For $t \sim U[0, 1]$,

$$x_t = (1 - t)x_0 + ty, \qquad u_t = y - x_0, \qquad \widehat{u}_t = v_{\vartheta_A}(x_t, t, c_A). \tag{3}$$

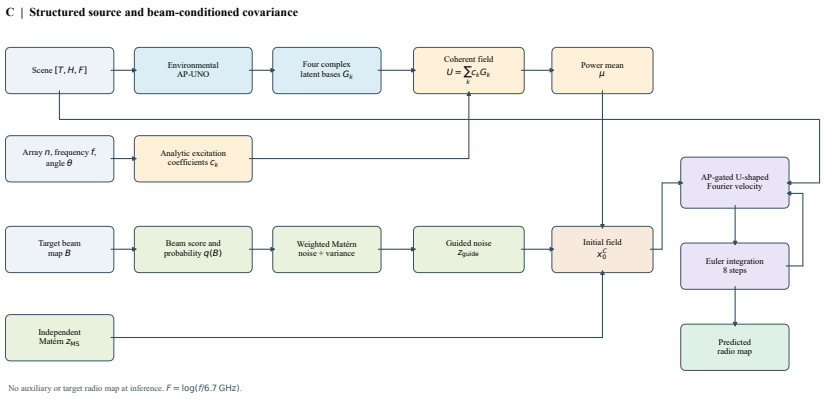


Figure 2: Variant C: environmental features and analytic excitation form a source mean, while the beam map modulates the source covariance.

The velocity network is a U-shaped multiscale Fourier operator. A condition encoder supplies multiresolution features; each operator stage combines conditioned attention, a truncated spectral convolution, a local linear branch, time embedding, and a residual connection. In an AP-gated stage, for retained Fourier mode k, the learned gate and spectral update are

$$g_\vartheta(k) = [1 + 0.5 \tanh a_\vartheta(k)] \exp\big(i \tfrac{\pi}{4} \tanh b_\vartheta(k)\big)\,, \tag{4}$$

$$\widehat{s}_o(k) = g_{\vartheta,o}(k) \sum_i W_{io}(k) \widehat{h}_i(k). \tag{5}$$

The gate networks receive normalized spectral features of the state and condition, their relative phase features, mode coordinates, and the time embedding. The gate is set to one on the implementation's excluded real-FFT boundary columns. This is modulation of latent Fourier features, rather than an estimate of the physical electromagnetic phase.

1.2 Variant C: Structured Source and Beam-Conditioned Noise

C separates the direct beam-map input from the velocity condition. It uses $c_C = [T, H, F, 0, 0, 0]$, where $F = \log(f/f_{\mathrm{ref}})$ is broadcast spatially and $f_{\mathrm{ref}} = 6.7$ GHz. An environmental AP-gated U-shaped Fourier network maps (T, H, F) to four complex latent basis maps $G_k(H, T, F; p)$, $k = 1, \ldots, 4$. These bases are learned from power-map supervision and are not identified as measured physical paths.

For an $n \times n$ array, let r_m be the implemented half-wavelength element offset and $w_m(n, \theta)$ the normalized steering weight. For the four fixed basis directions

$\varphi_k \in \{-30^\circ, -10^\circ, 10^\circ, 30^\circ\}$, the analytic excitation coefficient is

$$d_k = (\cos\varphi_k, \sin\varphi_k, 0), \tag{6}$$

$$c_k(a) = \sum_{m=1}^{n^2} \overline{w_m(n,\theta)}\,\sqrt{10^{g_k/10}}\exp\!\left(i\frac{2\pi f}{c_0} r_m^{\mathsf{T}} d_k\right), \tag{7}$$

$$g_k = 8 - \min\!\left\{12(\varphi_k/65^\circ)^2, 30\right\}. \tag{8}$$

The structured field and source mean are computed as

$$U(p) = \sum_{k=1}^{4} c_k(a) G_k(p), \qquad \mu(p) = \frac{10\log_{10}\!\left[(\lambda_f/4\pi)^2|U(p)|^2 + 10^{-30}\right] + 300}{300}, \tag{9}$$

where $\lambda_f = c_0/f$ and the implementation scales each complex network output by $1/100$ when forming G_k.

The normalized beam map B also constructs a distribution over model-grid positions. Let N be the number of grid cells and s_g the fixed training set 95th percentile of positive beam-gradient magnitudes. With $\alpha = \mathrm{softplus}(a_0)$ and $\beta = \mathrm{softplus}(b_0)$, define

$$d(p) = \mathrm{clip}\!\left(\frac{\|\nabla B(p)\|}{\max(s_g, 10^{-6})}, 0, 1\right), \tag{10}$$

$$q(p) = 0.8\,\frac{\exp[\alpha B(p) + \beta d(p)]}{\sum_u \exp[\alpha B(u) + \beta d(u)]} + \frac{0.2}{N}, \quad \rho(p) = Nq(p). \tag{11}$$

The scalars a_0, b_0 are trainable. Centered spatial differences are used inside the grid and one-sided differences at its boundary. On the circulant embedding, ρ is padded with one outside the model grid. Let $\mathcal{G}_\ell$ be the linear Matérn square-root filter, ψ_ℓ its convolution kernel, and ϵ_ℓ independent white Gaussian fields. For an embedding-grid location p, the guided field is

$$\widetilde{z}(p) = \frac{1}{\sqrt{3}} \sum_\ell \left[\mathcal{G}_\ell(\sqrt{\rho}\,\epsilon_\ell)\right](p), \tag{12}$$

$$D(p) = \frac{1}{3}\sum_\ell \sum_u \psi_\ell(p-u)^2 \rho(u), \qquad z_{\mathrm{guide}}(p) = \frac{\widetilde{z}(p)}{\sqrt{\max(D(p), 10^{-8})}}, \tag{13}$$

$$x_0^C(p) = \mu(p) + \sqrt{0.75}\, z_{\mathrm{MS}}(p) + 0.5\, z_{\mathrm{guide}}(p). \tag{14}$$

Fields are cropped to the model grid after embedding-grid filtering. D is the conditional pointwise variance computed from the filters, not a per-sample normalization. The beam affects the covariance of the source; this is internal noise modulation and does not acquire new radio measurements. C applies the same flow path as Eq. (3), with $v_{\vartheta_C}(x_t, t, c_C)$. The evolving state carries the structured source and beam-conditioned noise, so the final output is not constrained to be linear in the excitation.

1.3 Objectives and Inference

Let $e = M \odot (\widehat{u}_t - u_t)$ and Ω_M be the valid cells. The masked flow objective and two Haar-detail terms are

$$\mathcal{L}_{\mathrm{FM}} = \frac{\sum_p e(p)^2}{|\Omega_M|}, \tag{15}$$

$$\mathcal{L}_{\mathrm{H}}^{(j)} = \frac{\sum_{p,d} M_j(p,d)\,[\mathcal{H}_j(\mathrm{AvgPool}_2\, e)]_d(p)^2}{\sum_{p,d} M_j(p,d)}, \quad j \in \{1,2\}, \tag{16}$$

where $\mathcal{H}_j$ denotes the implemented three directional Haar details at level j, and M_j retains coefficients with complete valid support. The optimized objectives are

$$\mathcal{L}_A = \mathcal{L}_{\mathrm{FM}} + 0.05\mathcal{L}_{\mathrm{H}}^{(1)} + 0.05\mathcal{L}_{\mathrm{H}}^{(2)}, \tag{17}$$

$$\mathcal{L}_C = \mathcal{L}_A + \frac{0.1}{|\Omega_M|} \sum_{p \in \Omega_M} |\mu(p) - y(p)|. \tag{18}$$

Each objective uses its own source and velocity parameters; in C gradients also propagate through the structured mean and guided-source parameters. At inference, Euler integration with $N_{\mathrm{step}} = 8$ gives

$$x_{(j+1)/N_{\mathrm{step}}} = x_{j/N_{\mathrm{step}}} + \frac{1}{N_{\mathrm{step}}} v_\vartheta(x_{j/N_{\mathrm{step}}}, j/N_{\mathrm{step}}, c), \quad j = 0, \ldots, N_{\mathrm{step}} - 1. \tag{19}$$

The final field is resized from 256×256 to 128×128 and mapped to dB by $300x_1 - 300$. The present experiments use a fixed 6.7 GHz; the inputs and equations do not by themselves establish cross-frequency or cross-discretization generalization.